

ECON 712: Applied Econometrics

Lecture 14: Introduction to Instrumental Variables

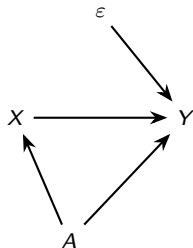
ECON712 Winter 2026

Instrumental Variables

- ▶ There are instances where we cannot solve endogeneity by adding controls, e.g.
 - ▶ Omitted variable bias: the omitted variable is not observed
 - ▶ Treatment Effects: treatment assignment does not satisfy the CIA assumption for any set of observable covariates
- ▶ We can solve this by finding an instrument Z that allows us to identify the causal effect of an endogenous variable X
 - ▶ An instrument only affects the outcome variable through X and does not correlate with unobserved variables affecting the outcome

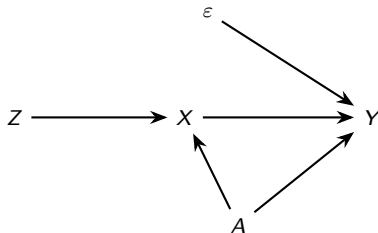
E.g. Omitted Variable Bias

- Suppose the true causal model is:



- This gives: $y = \beta_0 + \beta_1 x + \beta_2 A + \epsilon$

- An instrument z satisfies:



E.g. Omitted Variable Bias

- ▶ OLS on $y = \beta_0 + \beta_1 x + \eta$ (where $\eta = \beta_2 A + \epsilon$) gives a **biased and inconsistent** estimator.
- ▶ However: (i) $\text{Cov}(z, x) \neq 0$, (ii) $\text{Cov}(z, \eta) = 0$ (**exclusion restriction**), (iii) z only affects y through x
 - ▶ **Q:** If we observe z , can we estimate β_1 ? **YES**
- ▶ Using the reduced-form and first-stage:

$$\beta_1 = \frac{\text{Cov}(y, z)}{\text{Cov}(x, z)} = \frac{\gamma_1}{\alpha_1}$$

$$y = \gamma_0 + \gamma_1 z + u, \quad E[u|z] = 0 \quad (\text{Reduced-Form})$$

$$x = \alpha_0 + \alpha_1 z + \nu, \quad E[\nu|z] = 0 \quad (\text{First-Stage})$$

- ▶ Sample analogue:

$$\hat{\beta}_1^{IV} = \frac{\hat{\gamma}_1}{\hat{\alpha}_1}, \quad \hat{\gamma}_1 = \frac{\sum_i (y_i - \bar{y})(z_i - \bar{z})}{\sum_i (z_i - \bar{z})^2}, \quad \hat{\alpha}_1 = \frac{\sum_i (x_i - \bar{x})(z_i - \bar{z})}{\sum_i (z_i - \bar{z})^2}$$

- ▶ We **cannot** show unbiasedness in finite samples, but we can show consistency:

$$\text{plim}(\hat{\beta}_1^{IV} - \beta_1) = \frac{\text{plim} \frac{1}{n} \sum_i (z_i - \bar{z}) \eta_i}{\text{plim} \frac{1}{n} \sum_i (x_i - \bar{x})(z_i - \bar{z})} = \frac{\text{Corr}(z, \eta) \sigma_\eta}{\text{Corr}(x, z) \sigma_x} = 0$$

2SLS: General Vector Form

Setup. Partition regressors into exogenous controls $\mathbf{S}$ ($n \times k$) and one endogenous variable $\mathbf{x}$ ($n \times 1$):

$$\mathbf{y} = \mathbf{S}\boldsymbol{\delta} + \mathbf{x}\beta_1 + \boldsymbol{\eta} \equiv \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta}, \quad \mathbf{X} = [\mathbf{S} \ \mathbf{x}] \in \mathbb{R}^{n \times (k+1)}$$

Stack exogenous controls and j excluded instruments into the instrument matrix:

$$\mathbf{Z} = [\mathbf{S} \ \mathbf{Z}_2] \in \mathbb{R}^{n \times (k+j)}$$

Order condition (necessary for identification):

$$j \geq 1 \iff \underbrace{k+j}_{\# \text{ instruments}} \geq \underbrace{k+1}_{\# \text{ regressors}}$$

► $j = 1$: just-identified $j > 1$: overidentified $j < 1$: underidentified — **cannot estimate β_1**

2SLS estimator. Let $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top$:

$$\hat{\boldsymbol{\beta}}^{2SLS} = (\mathbf{X}^\top \mathbf{P}_Z \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{P}_Z \mathbf{y}$$

Intuitively, $\mathbf{P}_Z \mathbf{X}$ purges $\mathbf{x}$ of its endogenous component, keeping only the variation attributable to $\mathbf{Z}$.

2SLS: Just-Identified Case via FWL

Just-identified: $j = 1$, so $\mathbf{Z} = [\mathbf{S} \ \mathbf{z}]$ and $\dim \mathbf{Z} = \dim \mathbf{X}$.

Partitioned regression. Since $\mathbf{S}$ appears in both $\mathbf{X}$ and $\mathbf{Z}$, the projection $\mathbf{P}_Z$ leaves $\mathbf{S}$ unchanged: $\hat{\mathbf{X}} = \mathbf{P}_Z \mathbf{X} = [\mathbf{S} \ \hat{\mathbf{x}}]$.

Write the second stage as a partitioned regression:

$$\mathbf{y} = [\mathbf{S} \ \hat{\mathbf{x}}] \begin{pmatrix} \hat{\delta} \\ \hat{\beta}_1 \end{pmatrix} + \hat{\xi}$$

Apply FWL. Let $\mathbf{M}_S = \mathbf{I} - \mathbf{S}(\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{S}^\top$ partial out $\mathbf{S}$. Define residualised vectors:

$$\tilde{\mathbf{y}} = \mathbf{M}_S \mathbf{y}, \quad \tilde{\mathbf{x}} = \mathbf{M}_S \mathbf{x}, \quad \tilde{\mathbf{z}} = \mathbf{M}_S \mathbf{z}$$

By FWL, $\hat{\beta}_1^{2SLS}$ is the IV estimator in the partialled-out regression:

$$\hat{\beta}_1^{2SLS} = \frac{\tilde{\mathbf{z}}^\top \tilde{\mathbf{y}}}{\tilde{\mathbf{z}}^\top \tilde{\mathbf{x}}} = \frac{\hat{\gamma}_1}{\hat{\alpha}_1} = \hat{\beta}_1^{IV}$$

where $\hat{\gamma}_1 = \tilde{\mathbf{z}}^\top \tilde{\mathbf{y}} / \tilde{\mathbf{z}}^\top \tilde{\mathbf{z}}$ (reduced-form) and $\hat{\alpha}_1 = \tilde{\mathbf{z}}^\top \tilde{\mathbf{x}} / \tilde{\mathbf{z}}^\top \tilde{\mathbf{z}}$ (first-stage).

Key result: In the just-identified case **2SLS = IV = RF/FS ratio**. Since $\hat{\beta}_1^{2SLS}$ is a ratio of random variables, it has **no finite moments**: bias is undefined, and the distribution has Cauchy-like tails.

2SLS in Practice

► In practice:

1. Estimate first stage by OLS and obtain $\hat{\mathbf{x}} = \mathbf{P}_Z \mathbf{x}$:

$$\hat{x}_i = \hat{\alpha}_0 + \hat{\alpha}_1 s_{1i} + \cdots + \hat{\alpha}_k s_{ki} + \hat{\alpha}_{k+1} z_i$$

2. Substitute $\hat{\mathbf{x}}$ into the structural equation and estimate by OLS:

$$\mathbf{y} = \hat{\beta}_0^{2SLS} \mathbf{1} + \hat{\beta}_1^{2SLS} \hat{\mathbf{x}} + \mathbf{S}\hat{\delta} + \hat{\xi}$$

► Standard errors for $\hat{\beta}_1^{2SLS}$ are **larger** than OLS because:

1. Variation in $\hat{\mathbf{x}}$ is a subset of variation in $\mathbf{x}$
2. First-stage estimation uncertainty must be propagated

► **Do not** manually insert $\hat{x}_i$ into OLS — use `ivreg/ivreg2` in Stata for correct $\widehat{Se}(\hat{\beta}_1^{2SLS})$

► **2SLS is biased in small samples**, which is a particularly **IMPORTANT PROBLEM** when instruments are weak (**weak instrument problem**)

2SLS: General Case with Multiple Controls

- ▶ Model with k controls and j excluded instruments:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 s_{1i} + \cdots + \beta_{k+1} s_{ki} + u_i$$

- ▶ First stage:

$$\hat{x}_i = \hat{\alpha}_0 + \hat{\alpha}_1 s_{1i} + \cdots + \hat{\alpha}_k s_{ki} + \hat{\alpha}_{k+1} z_{1i} + \cdots + \hat{\alpha}_{k+j} z_{ji}$$

- ▶ Second stage:

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 \hat{x}_i + \hat{\beta}_2 s_{1i} + \cdots + \hat{\beta}_{k+1} s_{ki} + \hat{u}_i$$

- ▶ To test for weak instruments: F -test of *excluded* instruments $z_{1i}, \dots, z_{ji}$ in the first stage
 - ▶ Rule of thumb: $F > 10$ (Stock & Yogo); stronger threshold $F > 100$ sometimes recommended